

# Requirements Traceability Matrix Updates

## Requirement Summary Revisions

| ID        | Revised Requirement / Enhancement                                                                                                                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-0<br>1 | Users can view a home page that explains the purpose of RoomRadar QC and links to room search. Interview participants preferred using a dedicated website/application rather than manually searching CSV course files or external tools.                                                                                            |
| FR-0<br>2 | Users can search for available rooms by day, start time, end time, and building. Student interviews validated this as the core workflow because many students explained difficulty finding available classrooms quickly between classes. Building/location filtering was especially important to reduce walking time across campus. |
| FR-0<br>3 | The system validates that required search fields are provided. Students emphasized the importance of fast and simple room searching, making validation necessary to prevent incomplete searches.                                                                                                                                    |
| FR-0<br>4 | The system validates that start time is before end time. Interview feedback reinforced the need for accurate search results and proper room availability calculations.                                                                                                                                                              |
| FR-0<br>5 | Users can view available room results with room code, building, and searched availability time range information. Students wanted clear visibility into whether rooms were occupied and where the rooms were located before entering them.                                                                                          |
| FR-0<br>6 | Users can open a room detail page to view the room schedule for the selected day. Interview participants wanted to know future classroom occupancy so they could avoid interruptions from upcoming classes.                                                                                                                         |

- FR-0 7 Search state is preserved during the current browser session. Students explained that room searching often happens spontaneously during short breaks between classes, making quick navigation and preserved searches important.
- FR-0 8 Admin users can log in with Firebase authentication. Authentication protects administrative schedule upload functionality and prevents unauthorized users from modifying room data.
- FR-0 9 Admin-only pages are protected from unauthenticated users. Backend upload protection is implemented with Firebase token verification.
- FR-1 0 Admin users can log out and return to the public experience. Session control improves security and allows administrators to safely end authenticated sessions.
- FR-1 1 Admin users can choose semester season and year for uploaded data. Semester organization supports accurate room availability calculations across different academic terms.
- FR-1 2 Admin users can upload a CSV file containing course and room information. Interview participants were unaware that course schedule CSV data existed, reinforcing the usefulness of automating schedule imports into the system.
- FR-1 3 The backend stores semesters, rooms, and schedules in MySQL. Persistent storage allows the application to maintain accurate room availability and scheduling data over time.
- FR-1 4 The backend can search for rooms that do not conflict with scheduled classes. Interviews confirmed that avoiding occupied classrooms was one of the most important user needs.
- FR-1 5 The backend can return a room's schedule for a selected day. Students wanted to know whether a room would remain available before deciding to study there.

- FR-1 6 The backend partially handles database connection errors with API responses. Additional exception handling and reliability improvements can still be expanded further.
- NFR-01 The application provides a responsive web interface. Most interview participants preferred using the system on mobile phones while moving around campus, reinforcing the need for responsive design.
- NFR-02 The system uses CORS to allow the frontend to call backend APIs during development. Frontend/backend communication supports seamless development and testing workflows.
- NFR-03 Backend configuration can be controlled through environment variables. Environment-based configuration improves deployment flexibility and protects sensitive credentials.
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## Traceability Matrix Enhancements

| Req ID | Revised Interview-Based Traceability Addition                                                          |
|--------|--------------------------------------------------------------------------------------------------------|
| FR-01  | Students preferred a dedicated platform over manually searching room schedules or CSV files.           |
| FR-02  | Interviews validated building/location filtering because students wanted nearby rooms between classes. |
| FR-03  | Validation supports the fast and convenient workflow students requested during interviews.             |
| FR-04  | Accurate time validation prevents invalid searches and incorrect room availability calculations.       |

- FR-05 Students wanted room availability visibility before physically checking classrooms.
- FR-06 Students requested schedule visibility to avoid entering rooms shortly before another class begins.
- FR-07 Session persistence supports spontaneous room searching behavior between classes.
- FR-08 Firebase authentication secures administrative room and semester management functionality.
- FR-09 Backend-protected routes prevent unauthorized access to administrative schedule uploads.
- FR-10 Logout functionality supports proper session management and security practices.
- FR-11 Semester controls help separate schedule data across academic terms.
- FR-12 CSV uploads automate the process of importing classroom scheduling data.
- FR-13 MySQL persistence supports long-term room and schedule management.
- FR-14 Conflict-based searching directly addresses interview complaints about occupied rooms and wasted search time.
- FR-15 Room schedule visibility helps students determine whether they have enough uninterrupted study time.
- FR-16 Additional backend exception handling improvements may still be expanded further.

- NFR-01

Responsive UI design supports mobile usage while students move around campus.
- NFR-02

CORS configuration supports frontend/backend integration during development.
- NFR-03

Environment variable configuration improves deployment flexibility and security.

Test Case Mapping Revisions

| Test ID | Revised Description                                                                            |
|---------|------------------------------------------------------------------------------------------------|
| TC-11   | Submit admin upload with missing required frontend fields and verify validation errors appear. |
| TC-13   | Verify API behavior during backend request failures and invalid authentication scenarios.      |

Revised Known Gaps / Follow-Up Items

| Gap ID | Description                                                                                                                                | Impact                                                                                      | Suggested Fix                                                                         |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| GAP-01 | Frontend admin dashboard protection relies primarily on the presence of a stored token in localStorage before backend verification occurs. | A fake or expired token may temporarily expose the admin UI until backend validation fails. | Perform token validation against Firebase before rendering protected frontend routes. |

|        |                                                                                   |                                                                           |                                                                                 |
|--------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| GAP-02 | Frontend CSV upload instructions do not fully match backend-required CSV columns. | Users may upload improperly formatted files that fail backend processing. | Update frontend upload instructions to reflect all required backend CSV fields. |
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## Future Enhancement Ideas Mentioned During Interviews

- Live occupancy indicators
  - Reservation functionality
  - Notifications for room availability
  - Room images and classroom type information
  - Mobile-first UI improvements
  - Improved aesthetic design and interface polish
  - Live room availability updates
  - Student communication/collaboration features
  - Room usage counters/check-in systems
  - Future room planning and scheduling assistance
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## Final Project Justification Statement

Student interviews were primarily used to validate and refine the usability of the original RoomRadar QC requirements rather than replace the existing system requirements. The interview feedback strongly aligned **with the implemented features, particularly room searching, building filtering, schedule visibility, and mobile accessibility.**